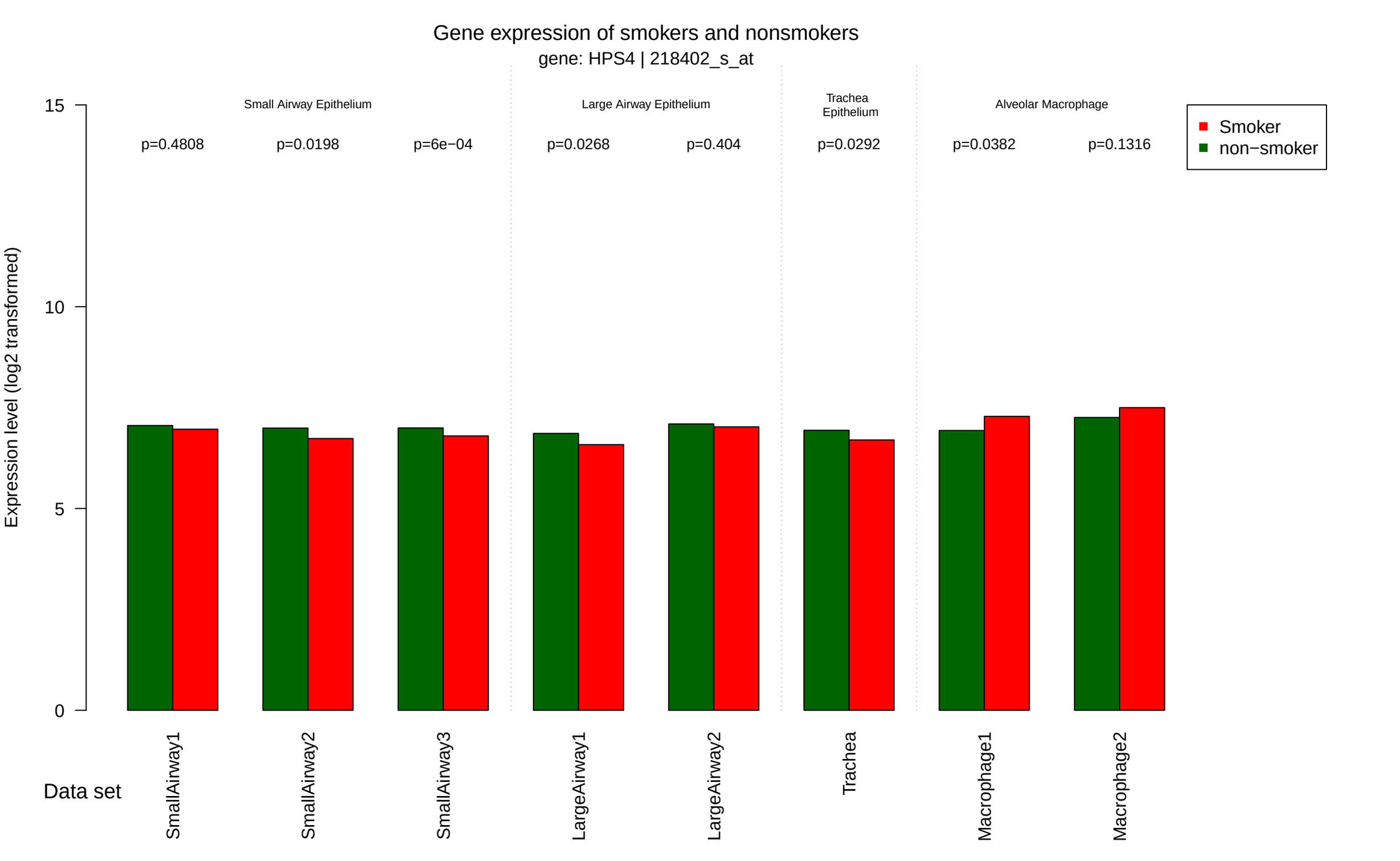
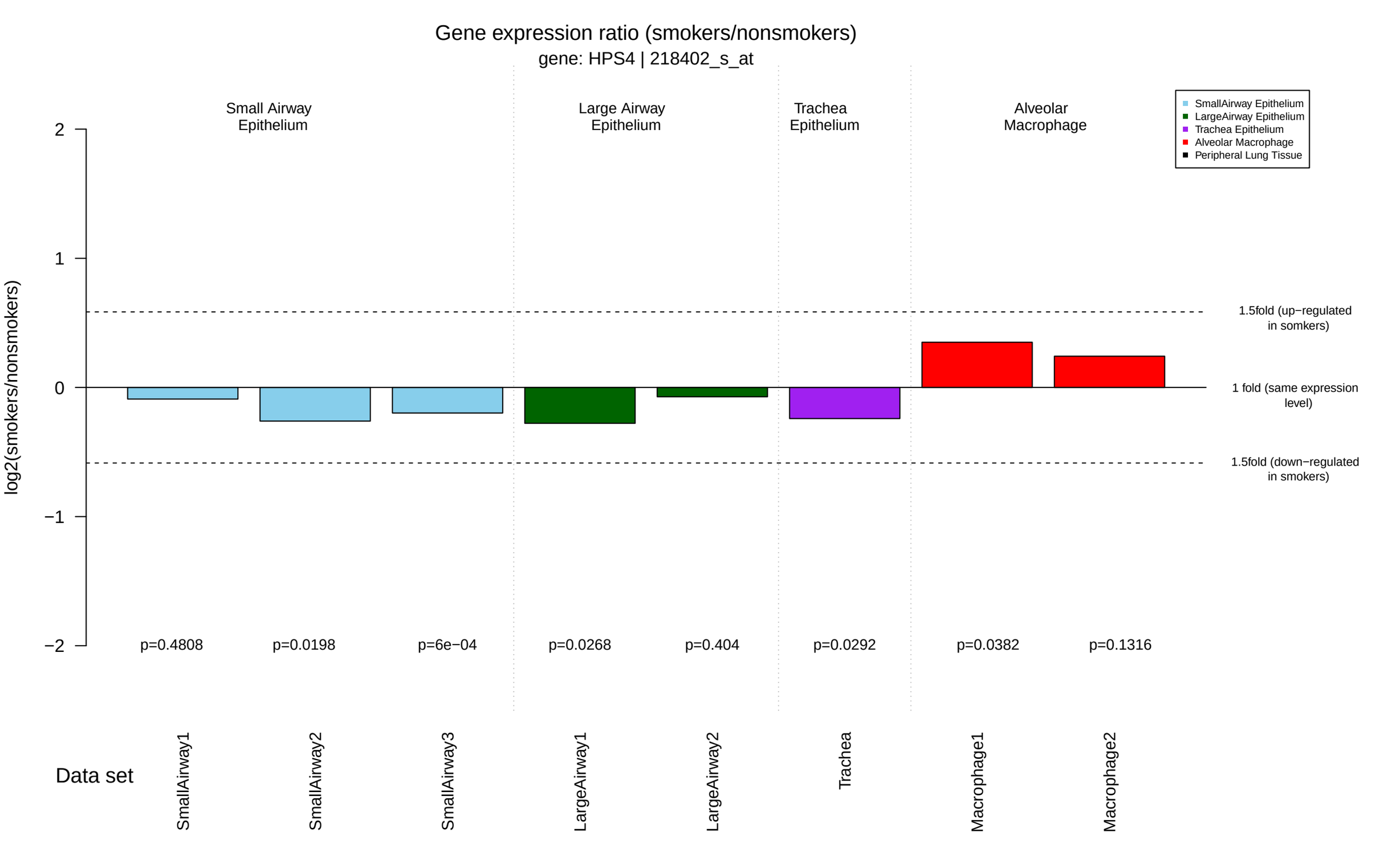
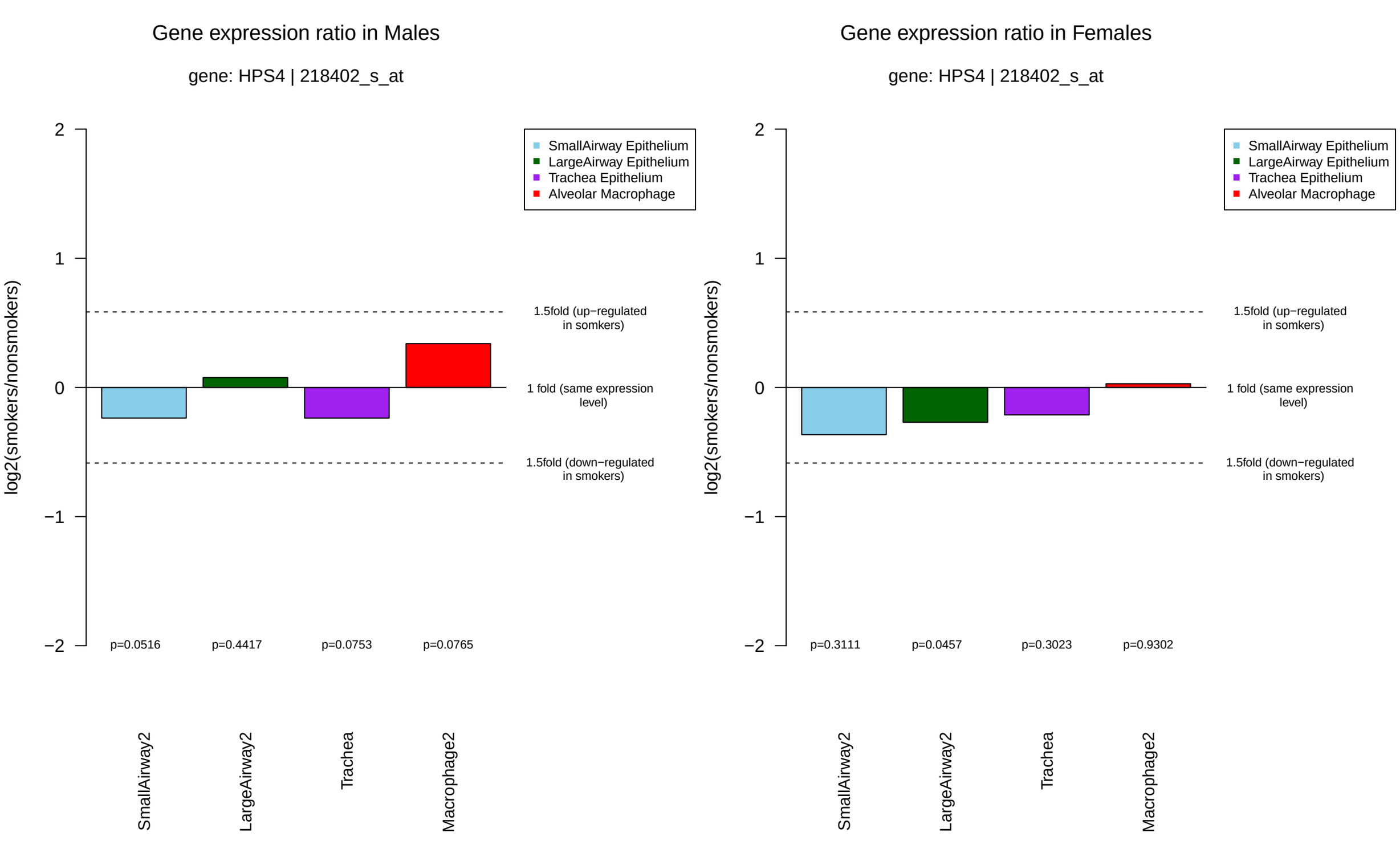
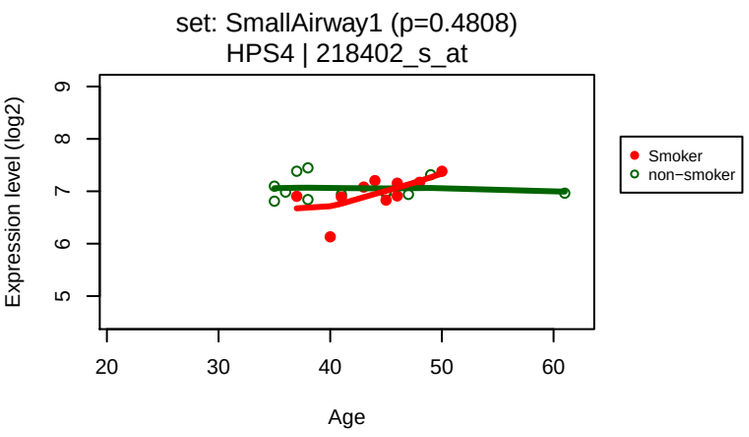
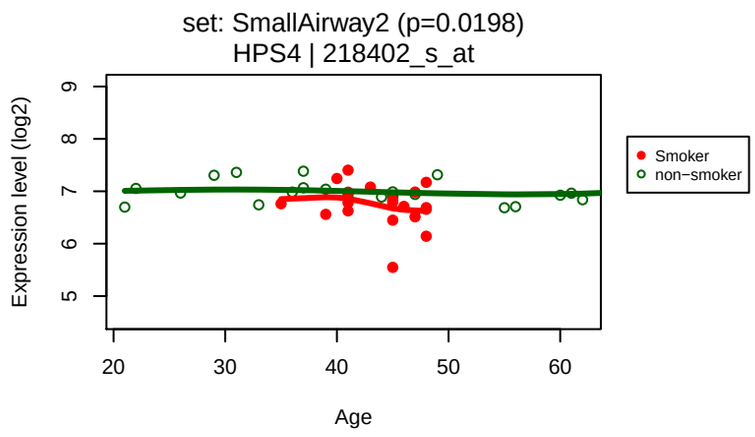
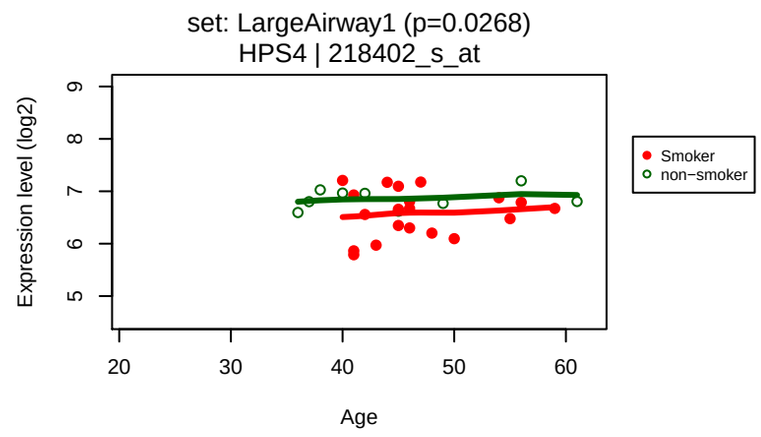

Gene expression of smokers and nonsmokers
gene: HPS4 | 218402_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4808
p=0.0198
p=6e−04
p=0.0268
p=0.404
p=0.0292
p=0.0382
p=0.1316
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

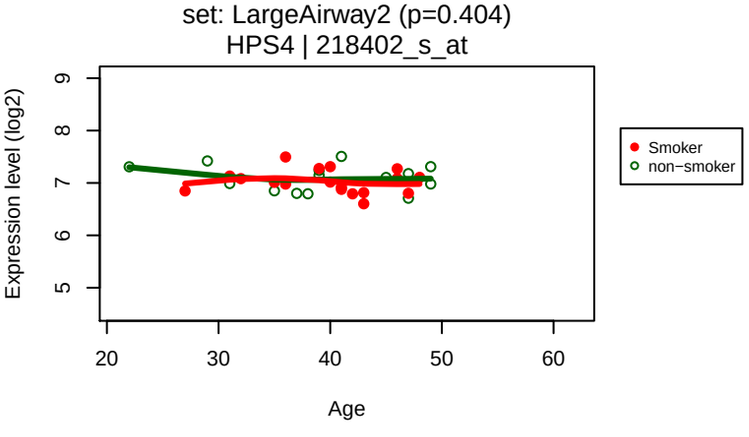

Gene expression ratio (smokers/nonsmokers)
gene: HPS4 | 218402_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4808
p=0.0198
p=6e−04
p=0.0268
p=0.404
p=0.0292
p=0.0382
p=0.1316
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

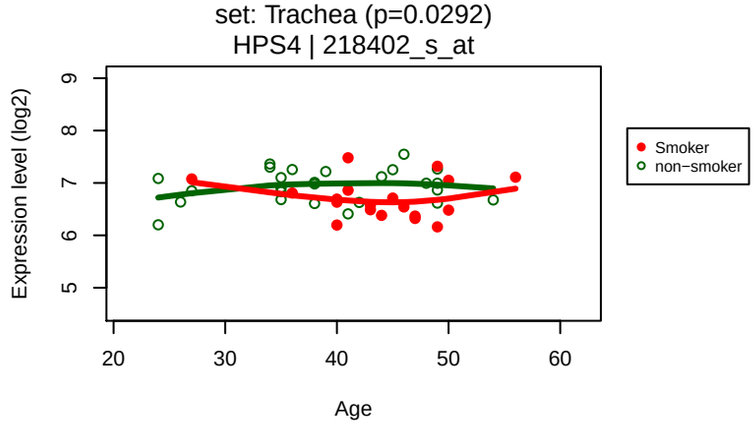

Gene expression ratio in Males
gene: HPS4 | 218402_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0516
p=0.4417
p=0.0753
p=0.0765
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPS4 | 218402_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3111
p=0.0457
p=0.3023
p=0.9302
SmallAirway2
LargeAirway2
Trachea
Macrophage2

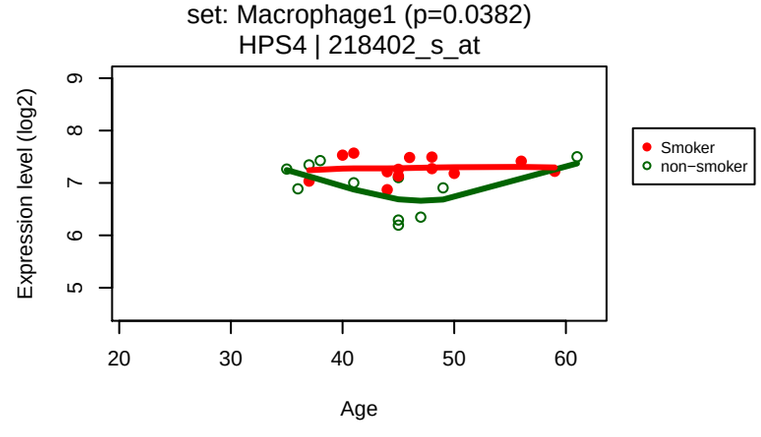

set: SmallAirway1 (p=0.4808)
HPS4 | 218402_s_at
Expression level (log2)
Age
Smoker
non−smoker

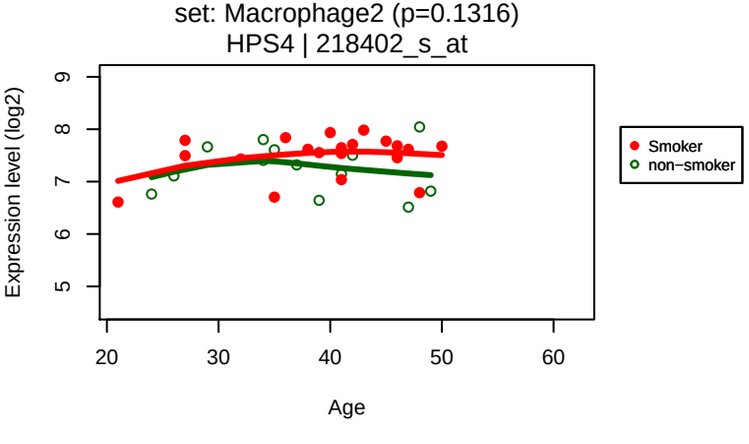

set: SmallAirway2 (p=0.0198)
HPS4 | 218402_s_at
Expression level (log2)
Age
Smoker
non−smoker

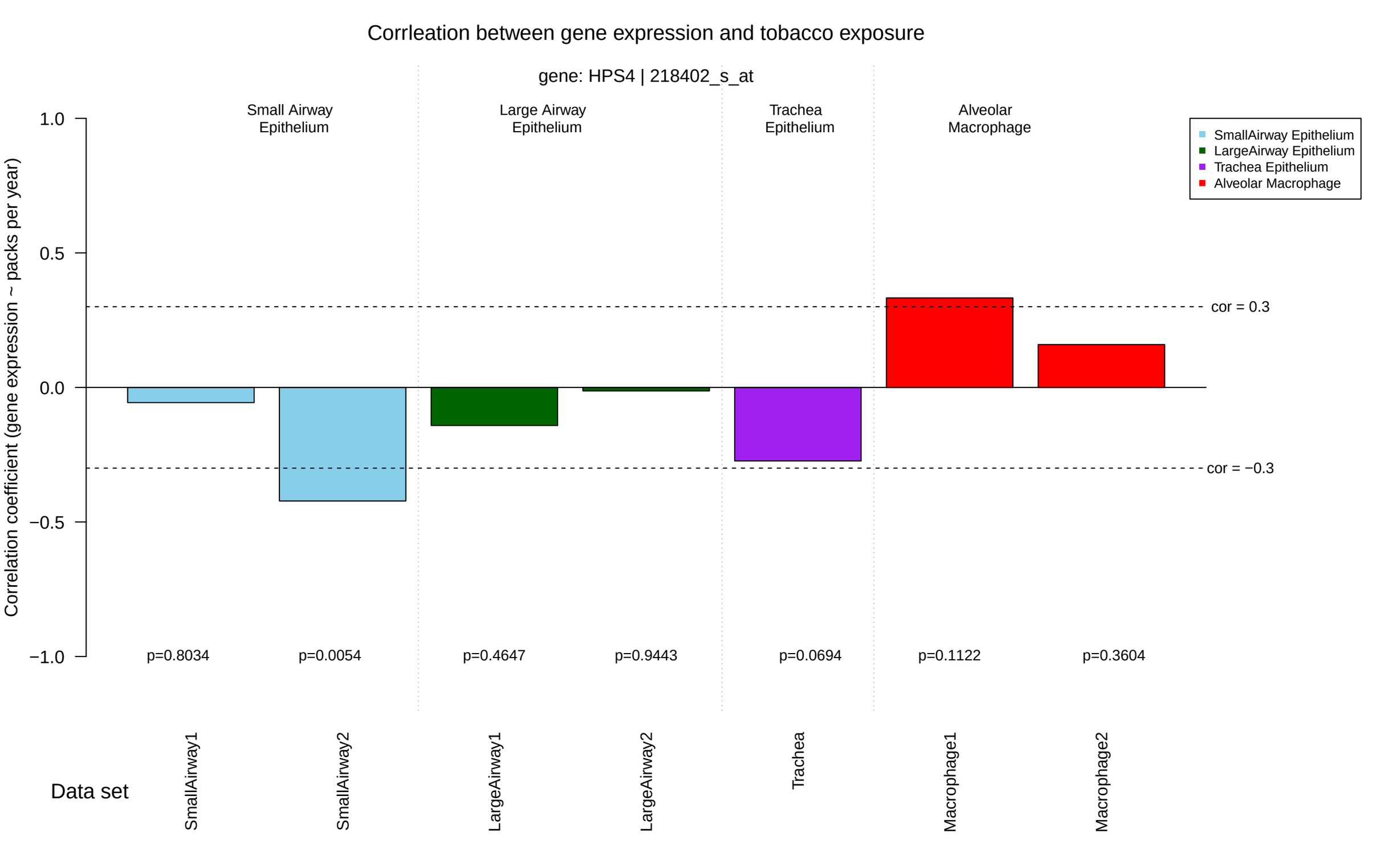

set: LargeAirway1 (p=0.0268)
HPS4 | 218402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.404)
HPS4 | 218402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0292)
HPS4 | 218402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0382)
HPS4 | 218402_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1316)
HPS4 | 218402_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPS4 | 218402_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8034
p=0.0054
p=0.4647
p=0.9443
p=0.0694
p=0.1122
p=0.3604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2